

Visibility-Aware Mobile Grasping in Dynamic Environments

Tianrun Hu, Anxing Xiao, David Hsu, Hanbo Zhang

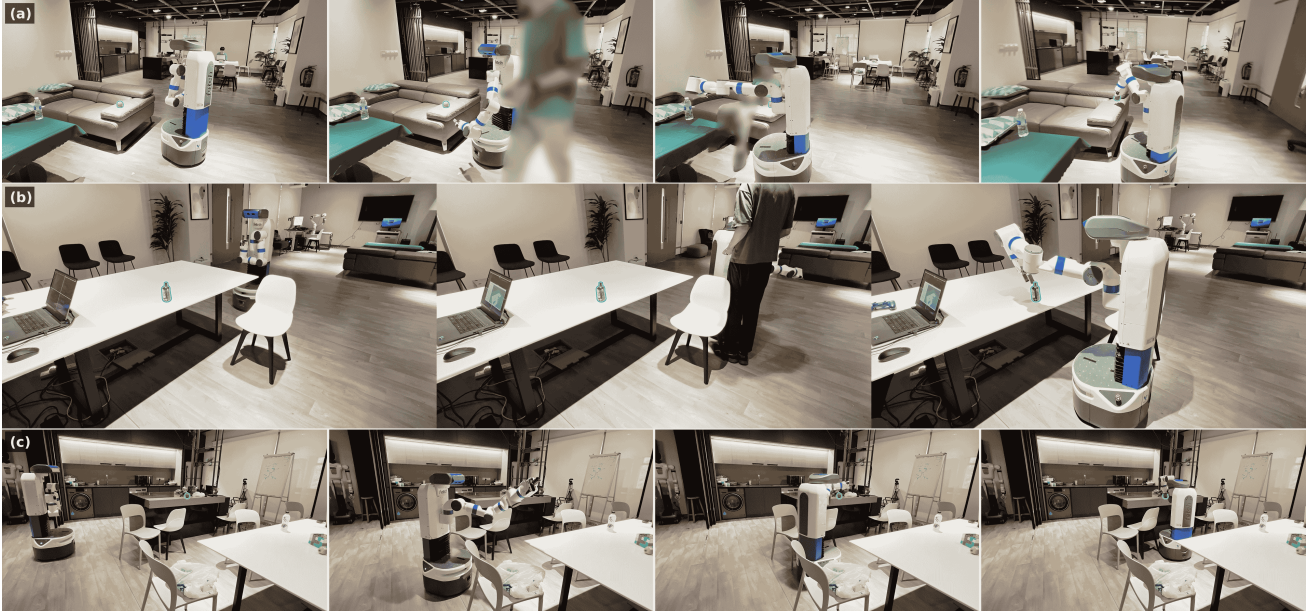


Fig. 1: Visibility-aware mobile grasping in the real world. (a) A moving person enters the robot’s route, and the robot reacts before grasping the apple. (b) The robot navigates around a person and chair to reach a bottle on a dining table. (c) The robot navigates through tightly spaced chairs before reaching an apple on the workstation. Faces are blurred for privacy.

Abstract—We consider mobile grasping in unknown and dynamic environments, where a robot must approach and grasp a target object using a camera with a limited field of view. This setting introduces two visibility constraints. Incomplete environmental observations can create collision risks along the whole-body trajectory, while occlusions can hinder object pose refinement during grasping. Previous systems often assume known or static environments and decouple perception from motion, leaving safety unresolved around unobserved obstacles. We present a unified system that enforces both visibility constraints at runtime through three core components. A velocity-aware gaze policy directs perception toward fast-moving regions of the planned swept volume, a hierarchical subgoal policy samples grasp and intermediate configurations that maintain target visibility, and a real-time whole-body planner replans over a receding horizon as the map updates. Across 400 randomized simulation scenarios and real-world trials on a Fetch mobile manipulator, our system achieves success rates of 70.4% in unknown static environments and 57.9% in dynamic environments, outperforming the decoupled navigation-and-manipulation strategy by 16.3 and 9.3 percentage points, respectively, while significantly reducing collisions. Project page: <https://visibility-aware-mobile-grasping.github.io/>.

I. INTRODUCTION

Bringing mobile robots from structured environments such as factories and warehouses to unstructured environments

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[1]–[3] such as homes, offices, and hospitals requires mobile manipulation beyond the common assumptions of a pre-built, perfect map and static obstacles. When the environment contains unknown clutter, such as a backpack left on the floor, or dynamic agents, such as a person walking by (Figure 1), the robot must actively use its onboard sensors to gather critical information while moving. In this paper, we focus on this challenging task regime. We propose a *visibility-aware mobile grasping system for unknown and dynamic environments that explicitly respects visibility constraints throughout the mobile grasping process*.

We formulate the problem as sampling over the constraint graph illustrated in Figure 2. Mobile grasping involves constraints on collision avoidance, kinematics, and grasping, which existing systems typically address under known maps. A limited field of view in unknown environments introduces two additional visibility constraints. The *collision visibility constraint* requires the maintained map to maximize coverage of the whole-body swept volume. Otherwise, a collision-free trajectory may still hit unobserved obstacles. The *object visibility constraint* requires the grasp configuration to keep the target in view for continued pose refinement. With a single camera, these constraints compete for gaze between the swept volume during motion and the target during grasping. We call this *visibility-aware mobile grasping*. Classical model-based [4], [5] and data-driven systems [2] struggle under such partial observability, while POMDP methods

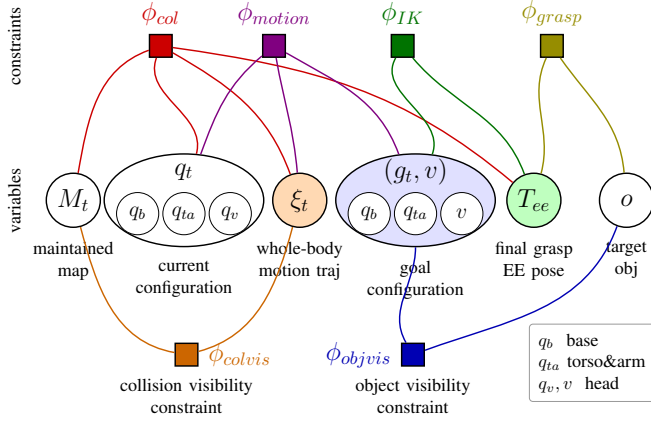


Fig. 2: **Constraint graph of visibility-aware mobile grasping.** Circles denote variables and squares denote constraints. Beyond collision ϕ_{col} , motion ϕ_{motion} , IK ϕ_{IK} , and grasping ϕ_{grasp} constraints, ϕ_{colvis} requires refreshed observation coverage of the swept volume of ξ_t , while ϕ_{objvis} requires goal g_t with head orientation v to keep target object o at p in view.

[6] are often intractable for high-dimensional whole-body manipulation under real-time requirements.

To address the visibility challenges while meeting practical deployment requirements, we present a unified system that enforces both constraints at runtime in a hierarchical and reactive manner. For the *collision visibility constraint*, we propose a velocity-aware active perception policy that continuously directs the camera toward the swept volume of the current trajectory, prioritizing fast-moving and imminent regions. For the *object visibility constraint*, we introduce a hierarchical subgoal policy that selects collision-free whole-body grasp and pre-grasp configurations while keeping the target in view. When manipulation is blocked or fails, the policy reactively falls back to more exploratory subgoals, such as observation poses. Finally, a real-time whole-body motion planner [7] and tracker replan within tens of milliseconds as the map updates. Together, these components turn dynamically changing visibility constraints into safe and reactive motion in the high-dimensional configuration space.

To evaluate our system and derive practical design principles for visibility-aware mobile grasping, we conduct an extensive study across 400 randomized simulation scenarios and real-world trials on a Fetch mobile manipulator. The experiments range from static environments with unknown layouts to dynamic scenes where obstacles cross the robot’s active route. Our system significantly outperforms the baselines in both unknown static and dynamic environments. The results suggest that interleaving whole-body planning with active perception enables mobile manipulators to operate more effectively in high-dimensional configuration spaces under real-world uncertainty. We further summarize the key design principles in the discussion section and provide a practical improvement direction for long-term open-world deployment of mobile manipulation systems.

II. RELATED WORK

A. Mobile manipulation systems.

Deployed mobile manipulation systems operate reliably in controlled settings such as retail environments and long-

term human-shared spaces [3], [8]–[11], but they depend on pre-built 3D maps and detect dynamic obstacles with 2D lidar, so they cannot react safely to changes in rich 3D structure. Reactive planners that address partial observability and dynamic obstacles [12], [13] are validated mostly in simplified simulations or low-dimensional configuration spaces, leaving real-time whole-body safety in cluttered, dynamic 3D scenes unaddressed. To manage the high dimensionality of mobile manipulation, prevailing pipelines decouple the problem through base placement optimization, selecting a fixed configuration that maximizes manipulability for a known goal [4], [5]. This assumes a nearly complete environment model and a goal fixed upfront. Our hierarchical subgoal generation extends this idea to partial observability, refining intermediate configurations incrementally as the map evolves.

B. Visibility-aware planning and active perception.

Visibility-aware planners constrain motion to previously observed free space [14], [15] under the assumption that observed regions remain valid in a static environment. Joint optimization of gaze and motion [16] maintains visibility of a fixed manipulation goal rather than one refined online. Active perception takes the complementary strategy of selecting viewpoints to maximize information gain. Frontier-based exploration builds complete maps [17], next-best-view planning optimizes viewpoints for grasping [18], [19], and recent work couples perception with manipulation to reduce occlusion [20], [21]. These methods separate viewpoint selection from motion execution and do not reason about swept-volume safety during replanning. Unlike them, our gaze policy weights observation by planned velocity over the robot’s future swept volume, so perception serves collision monitoring and goal refinement simultaneously. Real-time replanning itself builds on advances in reactive control [22], optimization-based [23] and sampling-based [24] planning, and vectorized sampling [7].

C. Learning-based mobile manipulation.

Modular systems ground language instructions into learned environment representations or pre-trained skills [1], [2], [25], [26]. Reinforcement learning trains end-to-end or modular whole-body policies at scale [27], [28]. Vision-language-action and embodied foundation models [29]–[32] add semantic breadth but degrade on unseen layouts and objects. These policies act without explicit geometric reasoning about collision risk and remain sensitive to distributional shift. Our system offers a complementary model-based alternative that reasons explicitly about visibility and swept volumes within a closed loop coupling perception and planning in dynamic, unstructured environments.

III. PROBLEM FORMULATION

We consider mobile grasping in unknown, dynamic environments using a single RGB-D head camera with limited field of view. The robot configuration is $q_t = (q_{r,t}, q_{v,t}) \in$

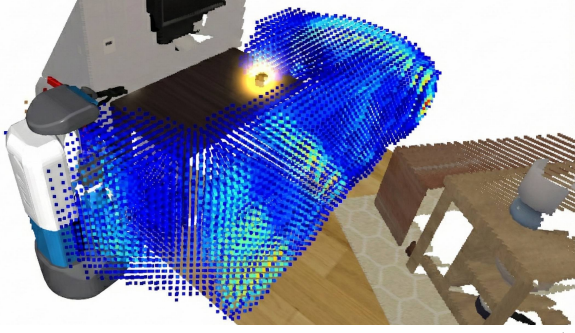


Fig. 4: **State-Dependent Gaze Optimization.** During planning ($\xi_t = \emptyset$), the importance field concentrates on the target region to enable grasp pose generation. During execution ($\xi_t \neq \emptyset$), the field shifts to the swept volume $V(\xi_t)$ (color-coded heatmap), weighted by velocity and temporal proximity to prioritize observation of collision-critical regions.

where $w_{safety}(x)$ focuses on near-future and high-velocity regions (Figure 4):

$$w_{safety}(x) = \sum_{i=0}^N \gamma^i \cdot w_d(x, q^{(i)}) \cdot \|\dot{q}^{(i)}\| \quad (4)$$

where i indexes a waypoint in ξ_t , γ^i decays with lookahead, $\|\dot{q}^{(i)}\|$ is the velocity at waypoint i , and w_d is a Gaussian centered at a moderate distance from the robot, where corrective action remains possible. The formulation extends [33] with velocity-aware, time-discounted weighting of the swept volume. We discretize the gaze space $\mathcal{A}_v$ into a pan-tilt grid and evaluate the objective exhaustively.

We maintain the map M_t as a point cloud updated via ray casting, where new observations add perceived surface points as obstacles while removing previously occupied space along each ray. This enables the robot to detect when dynamic obstacles have moved, while preserving occluded regions. The update pipeline includes ground plane removal, robot self-filtering, and voxel-based point merging.

B. Subgoal Policy π_g

Since the subgoal g_t depends on observed geometry, collision constraints, and robot reachability revealed as the map M_t changes, the subgoal policy π_g organizes grasping as a behavior tree with three fallback levels, namely grasp, pre-grasp, and observe goal sampling. It attempts the most direct level first and escalates to a more exploratory one when the current level fails, which also allows recovery from failures at runtime.

The policy first attempts to grasp without base motion, maximizing efficiency when the target is already within reach. We use a sample-then-verify strategy, in which a learning-based grasp network [34] generates candidate end-effector poses from the observed point cloud. These candidates are filtered by a pre-computed capability map [9]. We select N diverse poses via farthest point sampling and rank them by grasp confidence with a small penalty on the distance to the current end-effector pose. Each candidate is validated for whole-body inverse kinematics [35] feasibility and collision against the current map M_t , and the first valid configuration is selected as g_t . Execution reuses the receding-horizon loop, where the arm trajectory is monitored for

collisions and replanned as the map updates, followed by a short linear approach before closing the gripper.

When direct grasping fails, the robot must resample whole-body configurations that balance visibility and manipulability. Rather than solving a computationally expensive whole-body inverse kinematics problem, we factor the search over base pose, torso height, and end-effector pose into three conditional sampling distributions. The distribution p_{geo} samples collision-free base poses q_b from a 2D signed distance field that is updated in real-time from the point cloud. Given the sampled base pose, p_{torso} queries a pre-built torso height map to select h that maximizes the end-effector's reach to target p . Finally, p_{ee} samples end-effector poses T_{ee} in a sphere around the target, filtered by the capability map. Each sampled whole-body configuration (q_b, h, T_{ee}) is then validated via rejection sampling for inverse kinematics feasibility, collision, and self-occlusion, where the last check preserves visibility of the target. Both the capability map and the torso height map are pre-computed offline from large-scale sampled forward kinematics. The former inverts a reachability map to test whether an end-effector pose is reachable relative to the base, and the latter stores, for each end-effector position, the torso height that maximizes the number of feasible arm configurations.

When manipulation is blocked by occlusion or unknown geometry, the policy falls back to gathering observations that update the map M_t . We sample N base poses at a fixed safe standoff distance from the target, each paired with a robust arm configuration. These poses maintain visibility of the target p while minimizing displacement from the current body configuration $q_{r,t}$. If all three strategies fail, the policy signals failure to the behavior tree, triggering task termination.

C. Whole-Body Motion Policy π_r

Because M_t changes during execution, π_r must rapidly replan smooth, collision-free whole-body trajectories. Existing planners face a trade-off between speed and generality. VAMP [7] achieves real-time performance but assumes homogeneous configuration spaces, whereas non-holonomic mobile-manipulator planners [24] support heterogeneous base-arm spaces at higher computational cost. We combine their strengths to achieve both real-time performance and kinematic generality.

The key challenge is extending vectorized collision checking to non-holonomic steering. We therefore decompose planning hierarchically following [24]. The algorithm samples base placements using a non-holonomic kernel with Hybrid A* [36] for global guidance and Reeds-Shepp curves for local connections, then connects the corresponding arm configurations with RRT-Connect [37]. To respect their distinct kinematics, base and arm motions are discretized with Reeds-Shepp curves and linear interpolation, respectively, then synchronized at matching time steps. The resulting whole-body configurations support parallel SIMD validation [7], retaining vectorization speed without violating either kinematic structure.



Fig. 5: Qualitative results showing the execution sequence in simulation. From left to right: starting pose, observation stage, pre-grasp stage, and grasping stage. The top row shows the scene with the head camera view inset, and the bottom row shows the robot’s belief map.

After planning, randomized shortcutting [38], [39] removes unnecessary waypoints, B-spline fitting [40] smooths the trajectory, and dense re-interpolation validates it for collision-free execution. The resulting pipeline enables responsive replanning as M_t changes.

A whole-body velocity tracker executes the planned trajectory. At 20 Hz, it linearizes the non-holonomic base dynamics around the current heading and solves a one-step quadratic program in closed form, mapping the 11-D configuration error to a 10-D velocity command for the base, torso, and arm. This joint optimization keeps the three synchronized, which is critical because uncoordinated motion causes the end-effector to deviate. The tracker accepts trajectory updates during execution, enabling reactive replanning.

V. EXPERIMENTS

We evaluated our system across randomized simulation scenarios and real-world trials. Our experiments address three questions. First, how much does closing the loop between perception, subgoal reasoning, and whole-body planning improve over the prevailing decoupled designs? Second, how much of the improvement comes from velocity-aware active perception, and through which failure modes? Third, what structural limitations remain even with the full system?

A. Experimental Setup

a) Benchmark: For simulation experiments, we used ManiSkill3 [41] with ReplicaCAD scenes [42] for photo-realistic rendering and GPU-accelerated physics, modeling a Fetch mobile manipulator with accurate kinematics and dynamics. For real-world experiments, we used a Fetch mobile manipulator with ROS and localized the base with AMCL.

We evaluated three complementary scenarios. *Unknown static environments* tested perception and planning under visibility constraints with previously unseen layouts. *Dynamic environments* evaluated online perception and replanning against a moving obstacle. *Real robot deployment* validated real-world performance with localization and model errors. All scenarios were room scale (~ 5 -10m navigation range) and contained at least one collision-free path.

We generated 400 test scenarios (20 scenes $\times$ 20 objects) with procedural YCB object placement following [28]. Each

placement underwent validation: A drop-check verified object stability. An oracle grasp-check validated grasp feasibility using physics simulation following ACRONYM [43]. For each dynamic scenario, a collision-enabled obstacle with a 0.10×0.20 m footprint and 1.75 m height was instantiated on the active navigation route approximately 1.45 m ahead of the robot and moved transversely at 0.12 m/s throughout execution. The robot received no oracle obstacle state or trajectory. We accepted a placement only when the navigation mesh retained a collision-free detour to the target, and counted the dynamic interaction as valid only when replanning measurably changed the remaining route. Over the dynamic runs, placement succeeded in 97.5% of trials, replanning changed the route in 92.1%, and the mean closest approach to the obstacle was 0.81 m. For the learning baselines, we report Easy and Hard evaluation protocols. Easy follows the MS-HAB Pick initialization by placing the robot 0.6–1.6 m from and facing the target with a 200-step horizon. Hard uses exactly the same 400 tasks, robot starts, moving-obstacle settings, and physical success criterion as our benchmark with a 1,000-step horizon.

We tested at five representative indoor locations (dining table, kitchen counter, workstation, coffee table, sofa) with 40 configurations total: 20 in static unknown and 20 in dynamic scenarios with movable objects.

b) Evaluation Metrics: We evaluated system performance using grasp success rate as the primary metric. A trial succeeded if the robot lifted the target object above a height threshold (10 cm) and maintained a stable grasp, which means the object remained stable in the gripper for more than 2 seconds. We additionally report success weighted by path length (SPL), the ratio of the shortest collision-free path to the executed path over successful trials, to measure path efficiency. We categorized failures into three groups: *execution failures* (collision, grasp, IK) captured errors during physical interaction, *system limitations* (reachability, perception) reflected hardware and sensing constraints, and *planning failures* indicated that no valid subgoal or collision-free path was found within the planner’s budget of three 200 ms attempts.

c) Baselines and Ablations: We compare our full system against seven baselines and ablations spanning learning-based policies, conventional mobile manipulation designs, and controlled ablations.



Fig. 6: **Spatial Distribution of Success and Failure.** Green and red circles represent successful and failed grasping attempts, respectively.

SG-VLA SG-VLA [30] is a vision-language-action policy for mobile manipulation. As its original code is unavailable, we reimplemented its spatially grounded visual representation and trained it on Pick demonstrations from the MS-HAB pipeline.

RL (MS-HAB) This baseline uses the MS-HAB SAC Pick policy with its original training pipeline and configuration, directly predicting continuous whole-body actions from visual and robot-state observations.

Direct Grasping This baseline executes a fixed observation, pre-grasp, and grasp sequence with whole-body replanning, removing adaptive subgoal resampling and structured recovery to isolate the benefit of the subgoal policy π_g and behavior tree.

CapMap Placement This baseline selects a kinematically feasible base pose with our pre-grasp sampler and capability-map filtering [4], navigates there with the arm tucked, then switches to fixed-base arm planning.

Navigation-and-manipulation Following the dominant decoupled paradigm [4], [5], the robot navigates with its arm tucked to a base pose at a fixed target distance, then switches to fixed-base arm planning, replanning navigation after collisions.

Ours (w/o Velocity Weighting) This ablation removes velocity weighting from the gaze objective in Eq. 4, keeping only volume-based safety weights [33], to isolate velocity-aware active perception.

Ours (w/o Observation Stage) This ablation removes the dedicated observation stage and its structured behavior transitions, repeatedly replanning toward sampled pre-grasp configurations, to isolate the contribution of perception behavior and high-level task management.

B. Results and Analysis

a) *Qualitative Results:* Figure 5 shows a representative rollout. Notably, the sampled pre-grasp pose avoided self-occlusion and kept the target visible for the final grasp, reflecting the object visibility constraint in action.

b) *Comparison with Baselines:* Our integrated approach outperformed navigation-and-manipulation by 16.3 percentage points in unknown and 9.3 percentage points in dynamic environments (Table I). The decoupled design committed to a base placement before the robot could observe the target’s surroundings, which frequently produced unreachable configurations and doubled planning failures,

while its fixed-base arm planning struggled in constrained spaces and collided more often. Removing the subgoal policy entirely (Direct Grasping) caused the largest drop of 43.2 and 41.8 percentage points. Without the behavior tree, the system could not fall back to repositioning or observation gathering when grasp sampling failed, and planning failures grew by 3.3–5.3 $\times$. Even with the oracle Easy initialization, the learning baselines achieved low success, and SG-VLA remained near zero. Their failures were dominated by collisions, rising further under the Hard initialization, consistent with acting on visual input without an explicit model of collision geometry. Neither policy showed signs of seeking out the regions that mattered for safe motion, such as the space it was about to enter. Such behavior hardly emerges from the learning signal alone, so visibility needs to be treated as an explicit objective.

c) *Ablation Analysis:* Velocity weighting contributed through collision safety. Removing it more than doubled collisions in static environments and raised them by a third in dynamic ones while other failure categories changed little, so velocity-aware perception improves success almost entirely by avoiding collisions. The observation stage mattered less, lowering success by 2.1 and 1.6 percentage points, a difference not statistically resolvable over five runs (Welch’s t -test, $p = 0.20$ static and $p = 0.28$ dynamic). Path efficiency separates the two designs more clearly, with the full system reaching 62.2% and 48.0% SPL against 54.7% and 45.8% for the ablation, so without the observation stage the robot succeeds through longer and less direct paths. Per-scene success rates ranged from 58–82% in static and 41–67% in dynamic environments with cross-scene standard deviations of 6.9% and 7.5%, reflecting layout and clutter diversity.

d) *Real-World Validation:* We deployed only the full system on the real robot, as baselines lacking active perception and reactive replanning risks damaging the robot and the environment. It achieved success rates comparable to simulation in both static (65.0% vs. 70.4%) and dynamic (55.0% vs. 57.9%) environments (Figure 7). Planning failures primarily came from noisy observations (e.g., floating points in the point cloud) caused by data synchronization issues between the localization system and depth sensors. Collision failures arose from ground plane filtering that occasionally removed points from low-lying objects like chair legs, and from higher execution noise where the robot got stuck due to friction, then suddenly accelerated, disrupting pose estimation. Perception failures occurred when segmentation or detection errors led to incorrect grasping. These factors accounted for the performance gap between simulation and real-world.

e) *Runtime:* Each control cycle took approximately 100 ms (map update 69 ms, gaze optimization 26 ms, collision checking 3 ms), extending to roughly 150 ms when replanning was triggered (motion planning 50–80 ms). Subgoal sampling (369 ms) and grasp detection (1.2 s) ran only at phase transitions, 3.6 and 1.3 times per task. Velocity-aware gaze added 26 ms per cycle, the only computational difference from Ours (w/o Velocity Weighting). The pre-

TABLE I: Simulation success rates and failure breakdowns in percent. Entries report the mean with the 95% confidence-interval half-width from Student’s t distribution over five runs as a subscript. Lower is better for all failure categories. For RL and SG-VLA, grasp failures combine no-contact and unstable-hold outcomes. *Inval.* denotes dynamic trials without a measurable route change and applies only to the dynamic condition. Dashes indicate unavailable or inapplicable categories.

Method	Unknown Static Environment								Unknown Dynamic Environment							
	Succ. $\uparrow$	Coll.	Grasp	IK	Reach.	Perc.	Plan.		Succ. $\uparrow$	Coll.	Grasp	IK	Reach.	Perc.	Plan.	Inval.
SG-VLA (Easy)	1.6 _{1.2}	78.8 _{1.9}	19.6 _{1.8}	—	—	—	—		1.2 _{0.8}	80.5 _{2.1}	18.3 _{1.9}	—	—	—	—	—
SG-VLA (Hard)	0.0 _{0.0}	94.8 _{0.0}	5.2 _{0.0}	—	—	—	—		0.0 _{0.0}	97.8 _{0.0}	2.2 _{0.0}	—	—	—	—	—
RL (MS-HAB, Easy)	26.9 _{2.7}	42.7 _{3.1}	30.4 _{2.2}	—	—	—	—		24.1 _{3.0}	44.9 _{2.4}	31.0 _{2.8}	—	—	—	—	—
RL (MS-HAB, Hard)	12.0 _{0.0}	80.8 _{0.0}	7.2 _{0.0}	—	—	—	—		9.5 _{0.0}	84.5 _{0.0}	6.0 _{0.0}	—	—	—	—	—
Direct Grasping	27.2 _{2.5}	2.5 _{0.8}	1.6 _{1.0}	5.2 _{0.8}	2.5 _{0.1}	4.9 _{1.2}	56.3 _{3.2}		16.1 _{1.1}	14.8 _{1.9}	1.4 _{0.5}	1.4 _{0.3}	2.5 _{0.0}	4.0 _{0.8}	35.4 _{1.6}	24.4 _{1.2}
CapMap Placement	53.0 _{2.4}	9.7 _{1.0}	9.8 _{2.1}	0.0 _{0.0}	0.2 _{0.0}	7.5 _{0.9}	19.8 _{1.3}		49.7 _{0.9}	8.2 _{0.8}	9.1 _{1.4}	0.0 _{0.0}	0.0 _{0.0}	7.3 _{1.0}	16.3 _{1.0}	9.4 _{0.3}
Nav.-and-manip.	54.0 _{2.4}	7.7 _{1.1}	8.7 _{1.1}	0.0 _{0.0}	0.0 _{0.0}	7.6 _{1.0}	22.1 _{1.5}		48.6 _{3.4}	7.5 _{1.7}	8.7 _{1.7}	0.0 _{0.0}	0.0 _{0.0}	7.6 _{1.3}	18.1 _{1.7}	9.6 _{0.5}
w/o Vel. Weighting	62.2 _{1.8}	12.3 _{2.3}	8.4 _{0.6}	0.3 _{0.3}	3.1 _{0.3}	3.0 _{0.3}	10.4 _{0.3}		51.5 _{1.6}	18.1 _{2.4}	7.0 _{0.7}	0.6 _{0.3}	3.0 _{0.5}	2.7 _{0.1}	8.8 _{1.1}	8.3 _{1.2}
w/o Obs. Stage	68.3 _{3.7}	5.8 _{2.4}	7.2 _{2.9}	0.2 _{0.1}	1.3 _{0.3}	2.6 _{0.3}	14.6 _{1.3}		56.3 _{3.4}	13.2 _{4.4}	6.7 _{1.7}	0.3 _{0.3}	2.2 _{0.6}	2.5 _{1.1}	11.3 _{2.7}	7.5 _{1.0}
Ours (Full System)	70.3_{1.1}	5.0 _{1.5}	8.0 _{1.3}	0.5 _{0.3}	3.0 _{0.6}	2.5 _{0.9}	10.7 _{1.3}		57.9_{1.5}	13.9 _{3.4}	5.7 _{0.1}	0.1 _{0.1}	2.1 _{1.5}	2.6 _{0.6}	10.7 _{2.4}	7.0 _{1.2}

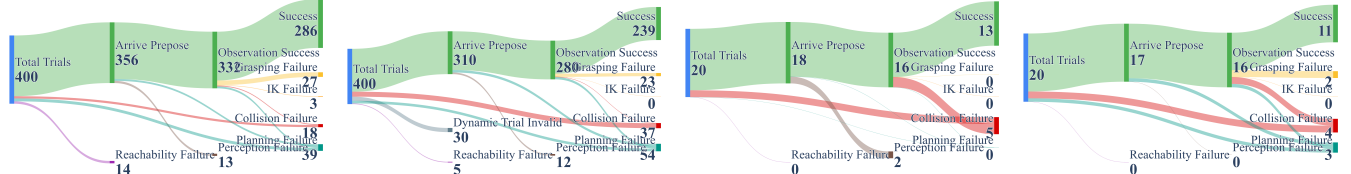


Fig. 7: Failure flow in simulation (left two, unknown static and dynamic) and on the real robot (right two, static and dynamic).

grasp sampler was empirically near complete, with success plateauing near 95% by 2,000 samples, motivating our 2,560-sample setting.

f) Failure Analysis: Figure 7 and Figure 6 show the failure flow and its spatial distribution on our benchmark. Failures concentrated on objects deep on tables, near walls, and in clutter, where feasible approach directions are scarce. The primary bottleneck was grasp detection. Among 46 manually labeled grasp failures, 43.5% were geometrically sound but kinematically infeasible grasps, 30.4% were misaligned poses, and 23.9% slipped during lift. The dominance of infeasible grasps reveals a domain shift from tabletop to mobile grasping, where the detector ranks candidates by geometric quality alone while reachability depends on base placement and environment constraints, making kinematically informed grasp scoring a clear direction.

Beyond this per-module attribution, the failure cases point to three structural limitations in the coupling between perception and planning. First, observed free space has limited temporal validity in dynamic scenes. As the ablation shows, velocity-aware gaze is effective once an obstacle has entered the map. Inspection of the residual collisions showed that space verified as free was later invalidated by obstacles arriving from adjacent unobserved regions before the gaze revisited it. Active perception should therefore anticipate where invalidation can originate, for example at occlusion boundaries near the planned route.

Second, reactive map maintenance retains stale evidence. The map cleared freed space only along observed rays, so removing points required re-observing the same region. Every dynamic obstacle naturally left such residue. Once the robot observed an obstacle and shifted its gaze, the obstacle departed while its points persisted in the map as floating obstacles, and regions the gaze had no reason to revisit kept them indefinitely, blocking valid paths and surfacing as planning failures. Removing this failure class requires a predictive environment model that propagates

dynamic evidence forward and decays unconfirmed points while preserving occluded static obstacles.

Third, the coupling between perception and planning remains unidirectional. The gaze policy conditioned on the planned trajectory, while the motion policy optimized collision cost in M_t without considering whether its swept volume could be observed. Observability is a property of the trajectory itself, determined by head reachability, self-occlusion, and motion direction, with backward motion as the extreme case where the forward-facing camera cannot cover the space being entered. Visibility should enter motion planning as a cost or constraint, extending joint gaze-motion optimization [16] from fixed goals to online refinement under an evolving map.

VI. CONCLUSION

This paper presented a unified framework for mobile grasping in dynamic, unknown environments under visibility constraints. By integrating velocity-aware active perception, hierarchical subgoal generation, and fast whole-body replanning, our system improves success rates by 16.3 and 9.3 percentage points over the navigation-and-manipulation method and achieves an overall 70.4% success rate in unknown static environments and 57.9% in dynamic environments. In particular, our method reduces collision rates effectively compared to baselines, benefiting from our closed-loop planning and active perception. These results demonstrate the importance of well-coordinated perception and planning for executing mobile grasping tasks in large-scale dynamic environments. Real-world experiments demonstrated strong generalization across diverse indoor settings. Future work follows the lessons identified in our analysis, including predictive environment models, active perception that anticipates unknown dynamics, visibility-aware motion planning, and kinematically informed grasp detection. The modular architecture naturally extends to multi-robot coordination, long-horizon or contact-rich manipulation tasks [44].

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